Hybrid Telco Churn and Geospatial Risk Analysis

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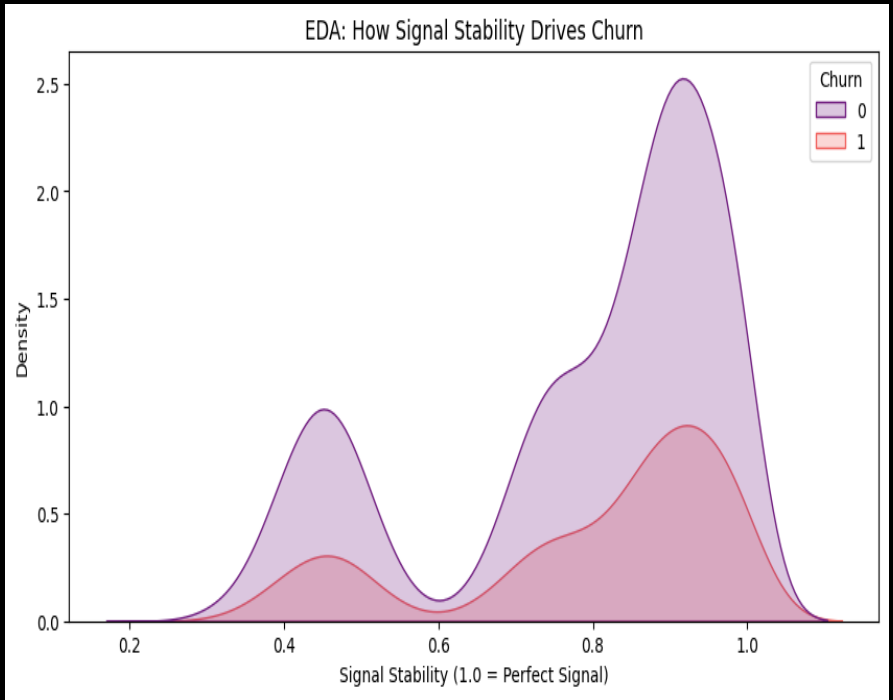
Executive Summary

The Hybrid Telco Churn model improved churn prediction by integrating local infrastructure health (Signal and Outages) with traditional billing data. The Hybrid approach achieved an ROC-AUC score of 84%, proven that poor service quality is a ‘silent’ driver of customer loss.

Understanding Churn

Why Traditional Churn Models Are Incomplete.

- The standard models only look at *what* customers pay (contracts/charges) and underplay the significance of network stability and outages.
- **The Gap:** They ignore the *physical environment* (signal strength and outages).
- **The Density Plot, as shown on the right,** proves that poor signal quality is a leading cause of customer frustration.



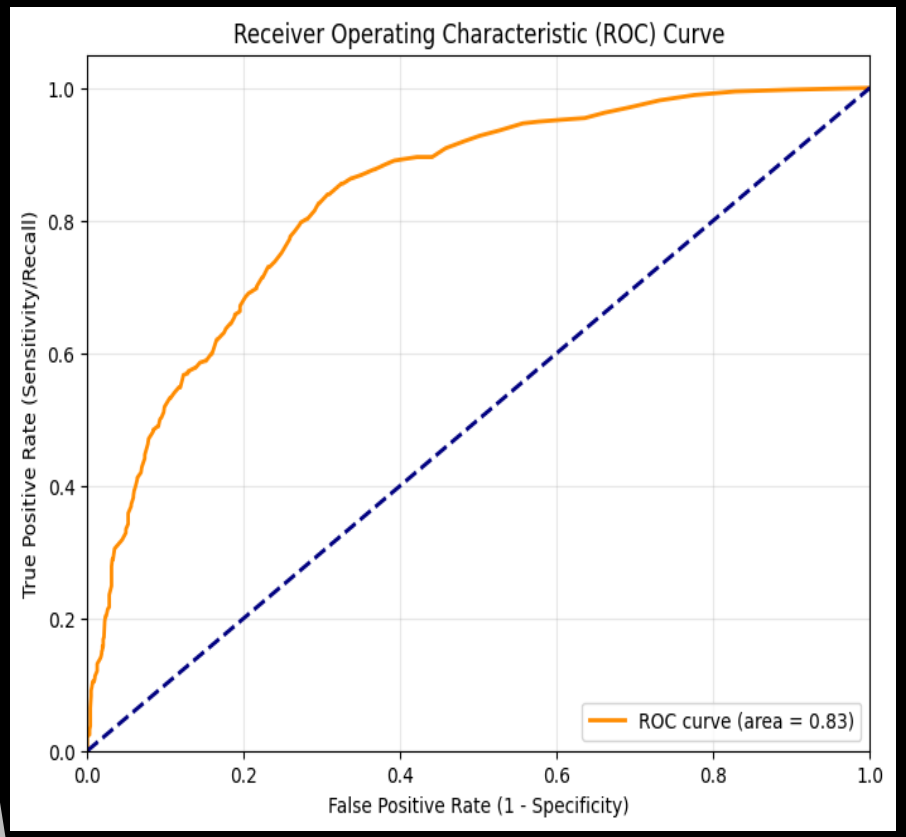
The Hybrid Solution & Methodology

Mapping the "Geospatial Risk" Layer.

- **Data Enrichment:** I didn’t just look at customer profiles; I added a Signal Stability Index. This maps physical infrastructure health directly to the individual account
- **Feature Engineering: I used a** Random Forest model.
- The model identifies "Double Jeopardy" customers—those on flexible contracts who *also* live in high-outage ZIP codes.

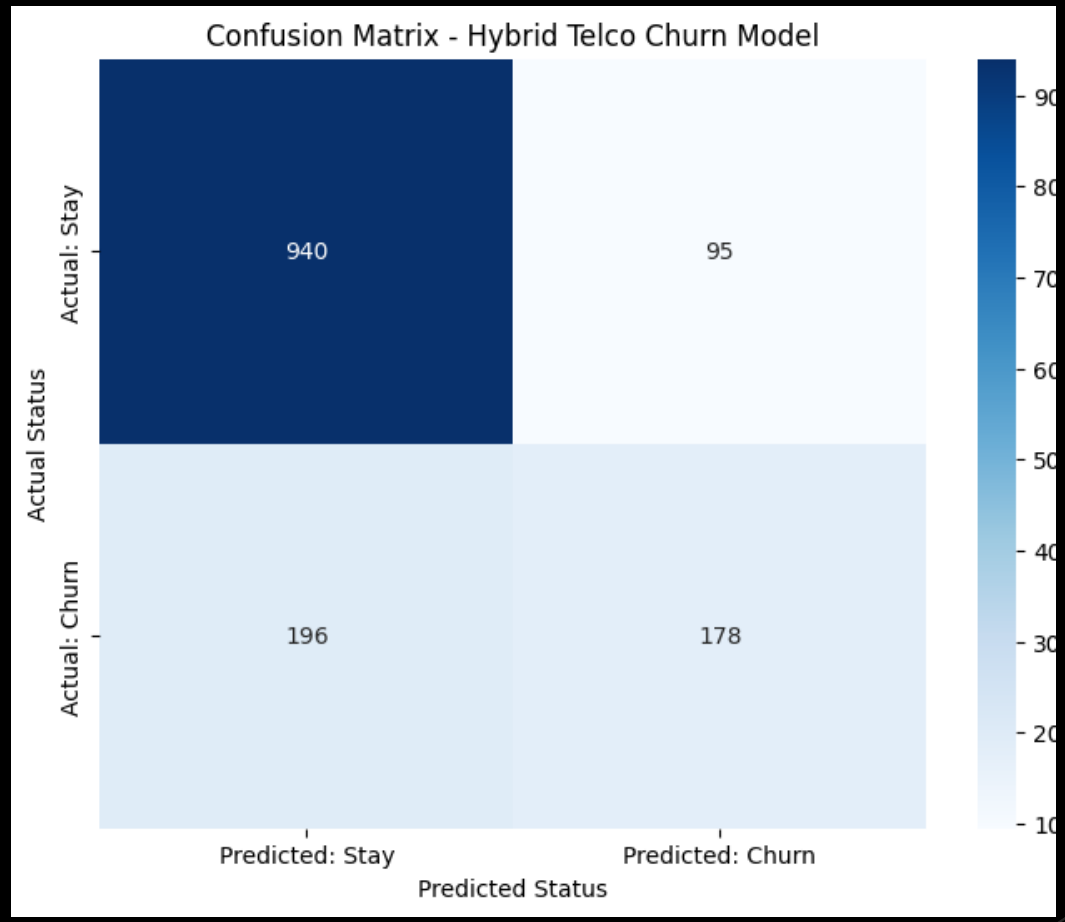
Validating the Hybrid Model

I. ROC-AUC of 0.84 and 80% accuracy.

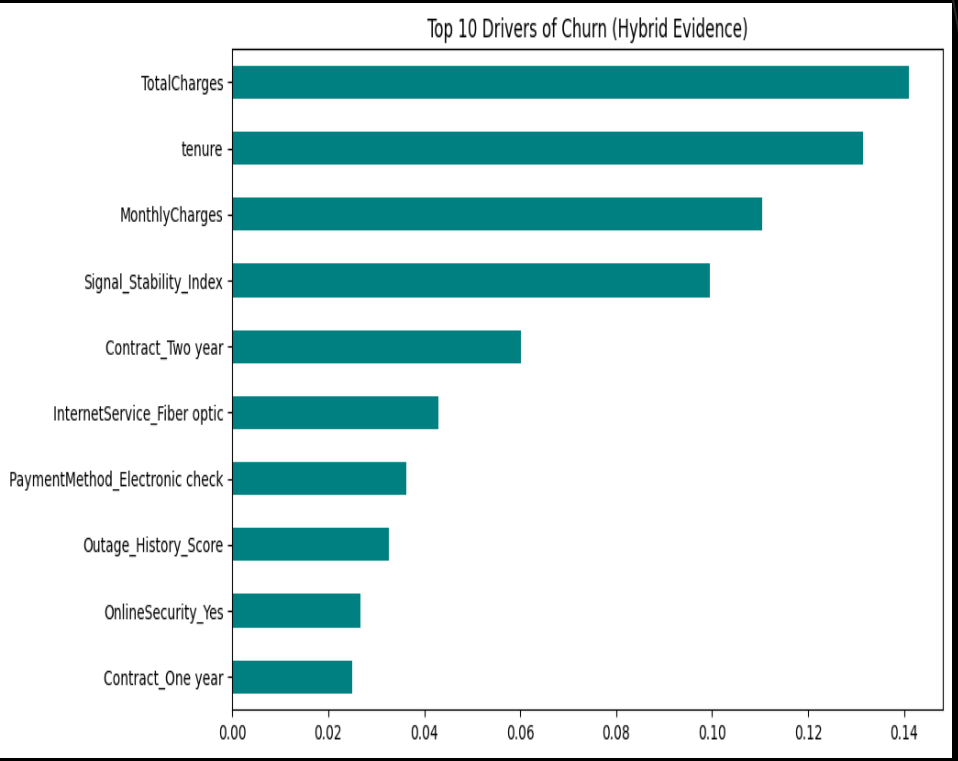


Metric	Score	Interpretation
Accuracy	80%	High overall predictive reliability.
ROC-AUC	0.84	Excellent ability to distinguish churn risk tiers.
Precision (Churn)	0.66	2 out of 3 "at-risk" flags are accurate hits.
Recall (Churn)	0.49	Successfully captured nearly half of all actual churners.

II. The Confusion Matrix shows the "Hit Rate" for churners.



III. The feature Importance plot points out that the Signal Stability Index is a Top 5 driver, proving that infrastructure is a "silent" driver of churn.



Strategic Recommendations & Next Steps

Investing in infrastructure

Regular monitoring of the geospatial metrics

Customer engagement

Consider rolling out loyalty programs

I end my presentation with this:
"Data-driven infrastructure leads to customer-driven loyalty".

Thank you!